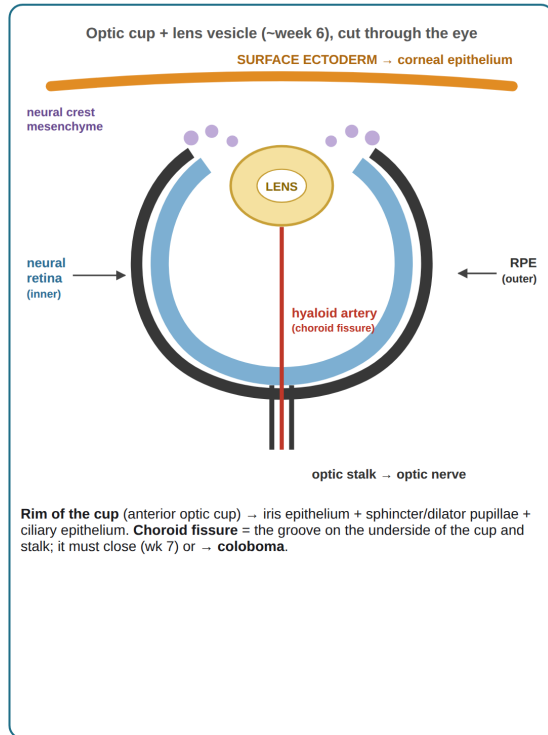


HNSS18 · Development of the Eye

Three sources build the eye: **neuroectoderm** (optic cup → retina, iris/ciliary epithelium), **surface ectoderm** (lens, corneal epithelium) and **neural crest** (corneal stroma, iris stroma, ciliary muscle).

~Day 17 Eye fields in the anterior neural plate (prosencephalon); express PAX6	Days 21–25 Optic grooves (sulci) invaginate	Day 28 Optic vesicles + optic stalks; vesicle touches surface ectoderm → lens placode induced	Days 30–36 Lens placode → lens pit → lens vesicle pinches off; vesicle invaginates → bilayered optic cup	Week 6 Choroid fissure forms inferiorly; carries the hyaloid artery	Week 7 Lens fibers fill the lens cavity; cornea differentiates; retina thickens; anterior chamber forms	Weeks 8–15 Eyelids form (end of embryonic period); iris (wk 8–9); ciliary processes (wk 15)	2nd → 3rd trimester Eyelids fuse at start of 2nd, reopen at start of 3rd; pupillary membrane regresses before birth
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Lens induction: a two-way conversation

- Optic vesicle** → induces → **surface ectoderm** → **lens placode** → **lens pit** → **lens vesicle** → detaches
- Reciprocal:** the lens placode makes the optic vesicle **invaginate** → **optic cup**
 - Pax6 + Sox2** make the surface ectoderm **competent** to respond to optic-vesicle **FGF/BMP** → lens placode
 - Primary lens fibers** = posterior cells elongate and fill the lens cavity (wk 7); **secondary fibers** = added from the equator throughout life (Maf)
 - The lens induces the cornea** **EXAM EMPHASIS**

Molecular control of the eye fields

- Eye fields arise in the **prosencephalon**, in line with the **diencephalon** → the eye (and optic nerve) is an outgrowth of the diencephalon **EXAM EMPHASIS**
- Eye fields express **PAX6** (master eye gene)
- SHH** from the anterior ventral midline **represses PAX6 in the midline** → splits one field into **two eyes**
- No SHH** → forebrain deficient, fields converge → **cyclopia**

Anterior segment & coverings

- Anterior chamber:** a space opens between the lens + iridopupillary membrane and the cornea (wk 7)
- Pupillary membrane** normally regresses before birth
- Iris stroma pigment** determines **eye color**
- Iris** (wk 8–9): anterior rim of the optic cup → epithelium + sphincter/dilator; NC mesenchyme → stroma
- Ciliary body** (wk 15): the two optic-cup layers fold → **ciliary processes**; NC → ciliary muscle + stroma. Ciliary processes secrete **aqueous humor** (→ IOP; drains via the **canal of Schlemm**); **suspensory ligaments** (zonules) + ciliary muscle → **accommodation**
- Eyelids:** form by end of embryonic period (wk 8) → **fuse** at start of 2nd trimester → **reopen** at start of 3rd. Folds = **surface ectoderm** (skin, conjunctiva) + **NC mesenchyme** core

Embryologic origins — where each part comes from

Neuroectoderm (optic cup / stalk)

Inner layer → neural retina
Outer layer → retinal pigment epithelium (RPE)
Only the neural retina has photoreceptors; the RPE has **none** **EXAM EMPHASIS**
Anterior rim → iris epithelium + sphincter & dilator pupillae + ciliary epithelium
Optic stalk + ganglion-cell axons → **optic nerve**

Surface ectoderm

Lens placode → lens vesicle → **lens**
Corneal epithelium (outer layer; cornea is **avascular** → transplants rarely rejected)
 Eyelid skin, lacrimal gland **STEP 1**

Neural crest (mesenchyme)

Corneal stroma + endothelium
Iris stroma
Ciliary muscle + stroma
Sclera + choroid (~wk 10)

Retina detail: as it develops, the **pigmented layer** thins while the **neural layer** thickens. **Retinal detachment** = separation between the neural retina (photoreceptor / outer nuclear layer) and the RPE — the old intraretinal space.
Hyaloid artery: proximal part persists → **central artery of the retina**; distal part degenerates → **hyaloid canal** left in the vitreous. **Optic chiasm** = crossing of **nasal** retinal fibers.
Superficial → **deep:** cornea → anterior chamber → iris → posterior chamber → lens → vitreous → neural retina → RPE → choroid → sclera.

Developmental abnormalities

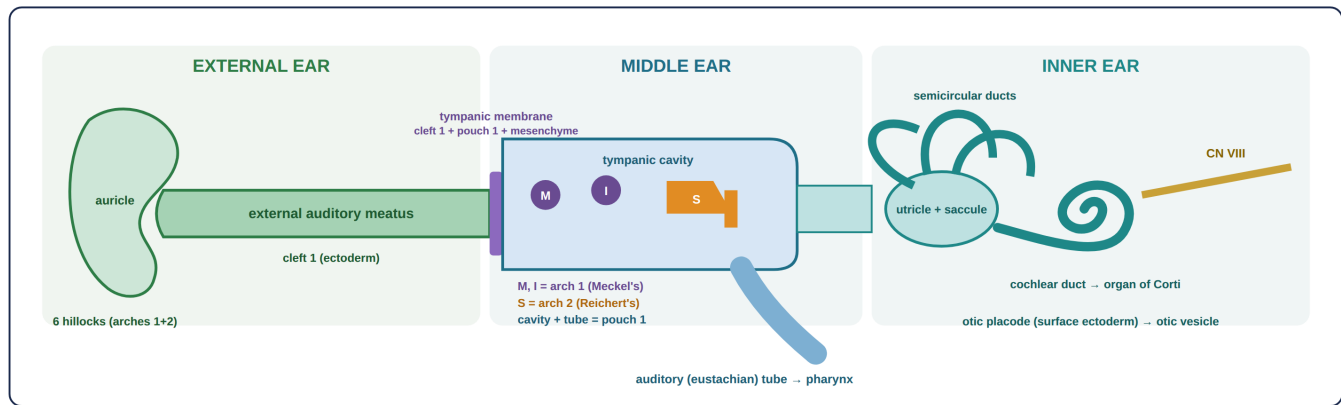
ABNORMALITY	EMBRYOLOGIC BASIS	CONSEQUENCE
Cyclopia	No SHH from the ventral midline → PAX6 not repressed → the two eye fields converge	Single midline eye; forebrain deficient. On the holoprosencephaly spectrum (SHH, trisomy 13, maternal diabetes, alcohol) STEP 1
Coloboma	Choroid fissure fails to close	Inferior "keyhole" gap in the iris; can involve retina, choroid, optic nerve
Persistent pupillary membrane	Pupillary membrane fails to regress before birth	Strands across the pupil; if it blocks the visual axis → deprivation amblyopia
Aniridia STEP 1	PAX6 mutation	Absent iris; part of WAGR (Wilms, Aniridia, GU, intellectual disability; 11p13)
Congenital cataract STEP 1	Lens injury in utero	Classic with congenital rubella (+ deafness, PDA)

Hook: "Cup, Cap, Crest." The optic **Cup** (neuroectoderm) makes the retina and the iris/ciliary *epithelium*; the surface ectoderm **Caps** it with the lens and corneal epithelium; **Crest** fills in the stroma and the ciliary muscle.

★ **USMLE high-yield:** Iris sphincter and dilator pupillae are **neuroectoderm** (one of the few muscles not from mesoderm). **Coloboma** = **failed choroid fissure closure** (inferior keyhole iris) — don't confuse with the **lower-eyelid coloboma of Treacher Collins** (arch 1 neural crest). **SHH loss** → **cyclopia/holoprosencephaly**. **PAX6** = **master eye gene** (aniridia).

HNSS18 · Development of the Ear

Three parts, three origins: **inner ear** from surface ectoderm (otic placode) · **middle ear** from pouch 1 + arches 1–2 · **external ear** from cleft 1 + arches 1–2.



External ear

1ST CLEFT + ARCHES 1–2

1st pharyngeal cleft → external auditory meatus

Auricle → 6 auricular hillocks (wk 5–6): 1–3 = arch 1, 4–6 = arch 2

Lobule is **not** from the hillocks

Ears start **caudal** (below the jaw); jaw growth moves them **up** and more vertical

External ear deformity → look for other pharyngeal-arch malformations **EXAM EMPHASIS**

Middle ear

1ST POUCH + ARCHES 1–2

1st pouch → tubotympanic recess → tympanic cavity + auditory tube (endoderm lines both)

Arch 1 (Meckel's) → malleus + incus

Arch 2 (Reichert's) → stapes

Tympanic membrane = pouch 1 + cleft 1 + intervening mesenchyme

Inner ear

SURFACE ECTODERM (OTIC PLACODE)

Otic placode → otic pit → **otic vesicle** (otocyst)

Vesicle → **membranous labyrinth**: utricle, semicircular ducts, saccule, cochlear duct, endolymphatic duct/sac

Placode-derived neuroblasts → **vestibular + spiral ganglia** → CN VIII

Inner ear timeline (human age)

~Day 22

Otic placode: thickened surface ectoderm beside the hindbrain

Days 26–28

Placode invaginates → **otic pit**, dorsal to the 2nd pharyngeal cleft

Day 29

Pit deepens and **pinches off** → **otic vesicle**

Day 30

Neuroblasts delaminate → **statoacoustic ganglion** between vesicle and neural tube

Day 36

Vesicle divides: **dorsal, utricular, saccular** portions

Week 9

Saccule outgrowth → **cochlear duct**; its epithelium → **organ of Corti**

What the otic vesicle becomes

OTIC VESICLE

→ Dorsal part → endolymphatic duct + sac

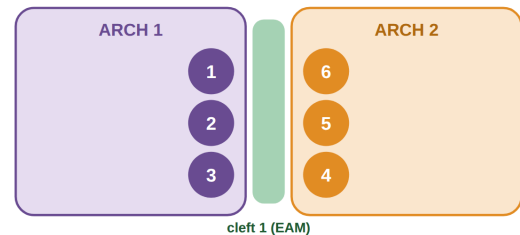
→ Utricular portion → utricle + semicircular ducts (balance)

→ Saccular portion → saccule + cochlear duct → organ of Corti (hearing)

→ Neuroblasts → statoacoustic ganglion → vestibular + spiral ganglia → CN VIII

Hook: “U for Up-and-around, S for Sound”: Utricle → semicircular ducts (rotation/balance); Saccule → cochlear duct (sound).

The auricle: six hillocks



- Hillocks 1–3 = arch 1 · 4–6 = arch 2 → fuse into the **auricle** (wk 5–6 → shaped by wk 9)
- Hillock 1 → **tragus**; hillock 6 → **antitragus**
- The **lobule** does **not** come from the hillocks
- Ears start **below the jaw**; jaw growth lifts them → poor jaw growth = **low-set ears**

Developmental abnormalities

ABNORMALITY	EMBRYOLOGIC BASIS	CONSEQUENCE
Low-set / malformed ears	Abnormal auricular hillocks (arches 1–2) or failed upward shift with jaw growth	Marker of arch or chromosomal syndromes (e.g., trisomies; check ear position on ultrasound)
Drug-induced ear anomalies LOWER PRIORITY	External ear is very sensitive to first-trimester drugs: streptomycin, salicylates, thalidomide	Pharyngeal and external ear malformations
Waardenburg syndrome	Autosomal dominant PAX3 (neural crest) defect affecting pigment cells	Congenital deafness + pigment changes (white forelock)
Genetic hair-cell loss LOWER PRIORITY	POU4F3 (hair-cell degeneration) · connexin 26 (supporting-cell gap junctions) · c-Kit (stria vascularis)	Sensorineural hearing loss
Treacher Collins STEP 1	Arch 1 neural crest (Tcof-1)	Malformed ossicles + external ear → conductive hearing loss
Congenital infection STEP 1	In-utero rubella or CMV	Sensorineural deafness (CMV = most common infectious cause; rubella + cataracts + PDA)

Hook: “Inner = placode, Middle = pouch, Outer = cleft.” Then the bones and auricle split by arch: **arch 1** → malleus, incus, hillocks 1–3 · **arch 2** → stapes, hillocks 4–6.

★ USMLE high-yield: Cleft 1 → external auditory meatus (only permanent cleft derivative); pouch 1 → tympanic cavity + auditory tube; tympanic membrane = all three germ layers. Inner ear (membranous labyrinth) = surface ectoderm (otic placode), **not** neural crest or arches. **Malleus/incus** = arch 1, **stapes** = arch 2. Low-set ears → think arch 1–2 or chromosomal (trisomy 18, 21) defects. **Inner ear develops first.** An **abnormal external ear** signals a pharyngeal-region defect → search for associated malformations **EXAM EMPHASIS**.